25901795 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCSTA100 - TIME SERIES METHODS AND THEIR APPLICATIONS

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **2** marks

1. What is white noise in time series analysis? [S] / [CO2]

   a). A random signal with no predictable pattern

   b). A deterministic trend in time series

   c). A seasonal fluctuation in a dataset

   d). A type of outlier in statistical data

2. What is the main reason for decomposing a time series? [U] / [CO1]

   a). To separate trend, seasonality, and randomness

   b). To remove missing values

   c). To change the scale of the data

d). To make the data more complex

3. What is the difference between a random walk and white noise? [S] / [CO2]

a). A random walk depends on past values, while white noise does not

b). White noise has a trend, but a random walk does not

c). A random walk has a constant mean and variance

d). White noise accumulates over time

4. What is a time series plot? [K] / [CO1]

a). A graphical representation of data points over time

b). A scatter plot with regression lines

c). A histogram showing frequency distributions

d). None of the above

5. If the residuals of a time series model show a pattern, which assumption is likely violated? [U] / [CO1]

a). No autocorrelation b). Homoscedasticity

c). Stationarity d). Normality

6. What does differencing help achieve in time series analysis? [S] / [CO1, CO2]

a). Makes the data non-stationary

b). Makes the series stationary by removing trends

c). Stabilizes the variance

d). Removes seasonal components

7. What is the range of autocorrelation coefficient? [I] / [CO1, CO3]

a). 0 and infinity b). 0 and 1

c). -1 and +1 d). None of the above

8. If points in a scatter plot are randomly scattered, what is the likely correlation? [I] / [CO3]

a). Positive correlation b). Negative correlation

c). Perfect correlation d). Weak correlation

9. In an AR(p) model, what does "p" represent? [S] / [CO1, CO2]

a). The number of lagged values used

b). The number of moving average terms

c). The number of seasonal components

d). The variance of the process

10. How can an AR model be converted into a stationary form if it's non-stationary? [I] / [CO1, CO3]

a). Differencing the series b). Adding more lags

c). Using logarithmic transformation d). Ignoring the first few observations

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **3** questions

Each question carries **5** marks

11. What is the main difference between cyclical and seasonal components in time series? Explain with an example. [An] / [CO1]

12. Differentiate deterministic time series from non-deterministic time series. [An] / [CO1]

13. What is the main idea behind the semi-average method of determining the secular trend? Explain. [S] / [CO1, CO2]

14. What is the difference between correlation and co-variance? [I] / [CO1, CO3]

15. How will you differentiate strict sense stationary process and weak stationary process? [S] / [CO2]

[5x3 = 15]